

Ball Launcher Velocity Equation

1 Problem Overview

This document derives the equation for the initial velocity v_0 required for a ball launcher to hit a target cup at a horizontal distance d , measured by a distance sensor. The arm rotates clockwise from the upper left quadrant (interpreted as ϕ between 90° and 180° , measured counterclockwise from the positive x-axis) toward the target on the right. The launch angle θ is a variable, with a conversion to the arm angle ϕ for motor/encoder programming.

2 Given Parameters

- Arm length: $r = 0.131$ m
- Shaft height: $h = 0.7112$ m (28 inches above ground)
- Target distance: d (variable, measured by sensor)
- Target height: $h_{\text{cup}} \approx 0$ m (ground level)
- Gravity: $g = 9.81 \text{ m s}^{-2}$
- Launch angle: $\theta \in [-90^\circ, 0^\circ]$ (variable, in radians)
- Arm angle: $\phi \in [90^\circ, 180^\circ]$, where $\theta = \phi - 180^\circ$

3 Launch Position

The ball is released at arm angle ϕ , measured counterclockwise from the positive x-axis (0° , to the right). The throwing angle is $\theta = \phi - 180^\circ$. The release coordinates are:

- $x_0 = r \cos \phi$ ($x_0 < 0$ for $\phi \in (90^\circ, 180^\circ]$)
- $y_0 = h + r \sin \phi$ ($y_0 > h$ for $\phi \in (90^\circ, 180^\circ]$)

4 Equation Derivation

The derivation computes v_0 as a function of d and θ , with the conversion from θ to ϕ .

4.1 Projectile Motion Equations

- Horizontal: $x(t) = x_0 + (v_0 \cos \theta)t$
- Vertical: $y(t) = y_0 + (v_0 \sin \theta)t - \frac{1}{2}gt^2$

The target is reached when $x = d$ and $y = 0$.

4.2 Step 1: Time of Flight

$$t = \frac{d - x_0}{v_0 \cos \theta}$$

Substitute $x_0 = r \cos \phi$, and since $\theta = \phi - 180^\circ$:

$$\cos \theta = \cos(\phi - 180^\circ) = -\cos \phi$$

$$t = \frac{d - r \cos \phi}{v_0(-\cos \phi)} = \frac{r \cos \phi - d}{v_0 \cos \phi}$$

4.3 Step 2: Vertical Equation

$$0 = y_0 + (v_0 \sin \theta)t - \frac{1}{2}gt^2$$

$$\sin \theta = \sin(\phi - 180^\circ) = -\sin \phi$$

$$0 = (h + r \sin \phi) - v_0 \sin \phi \cdot t - \frac{1}{2}gt^2$$

Substitute t :

$$0 = (h + r \sin \phi) - \sin \phi \left(\frac{r \cos \phi - d}{\cos \phi} \right) - \frac{1}{2}g \left(\frac{r \cos \phi - d}{v_0 \cos \phi} \right)^2$$

4.4 Step 3: Solve for Initial Velocity

$$v_0 = \sqrt{\frac{g(d - r \cos \phi)^2}{2 \cos^2 \phi \left[h + r \sin \phi - \sin \phi \left(r - \frac{d}{\cos \phi} \right) \right]}}$$

5 Example Calculations

5.1 Example 1: $d = 1.4 \text{ m}$, $\phi = 135^\circ$

- $\theta = 135^\circ - 180^\circ = -45^\circ$
- $x_0 = 0.131 \cos 135^\circ = 0.131 \cdot (-0.7071) \approx -0.0926 \text{ m}$
- $y_0 = 0.7112 + 0.131 \sin 135^\circ = 0.7112 + 0.131 \cdot 0.7071 \approx 0.8038 \text{ m} > 0.7112 \text{ m}$
- $v_0 \approx 3.0 \text{ m s}^{-1}$ (via numerical solution)

5.2 Example 2: $d = 1.2 \text{ m}$, $\phi = 120^\circ$

- $\theta = 120^\circ - 180^\circ = -60^\circ$
- $x_0 = 0.131 \cos 120^\circ = 0.131 \cdot (-0.5) = -0.0655 \text{ m}$
- $y_0 = 0.7112 + 0.131 \sin 120^\circ = 0.7112 + 0.131 \cdot 0.8660 \approx 0.8246 \text{ m} > 0.7112 \text{ m}$
- $v_0 \approx 2.7 \text{ m s}^{-1}$ (via numerical solution)

5.3 Example 3: $d = 1 \text{ m}$, $\phi = 150^\circ$

- $\theta = 150^\circ - 180^\circ = -30^\circ$
- $x_0 = 0.131 \cos 150^\circ = 0.131 \cdot (-0.8660) \approx -0.1134 \text{ m}$
- $y_0 = 0.7112 + 0.131 \sin 150^\circ = 0.7112 + 0.131 \cdot 0.5 \approx 0.7767 \text{ m} > 0.7112 \text{ m}$
- $v_0 \approx 2.3 \text{ m s}^{-1}$ (via numerical solution)

6 Notes

- The arm rotates clockwise from $\phi = 90^\circ$ to 180° , measured counterclockwise from 0° (*right*). $\phi \in [90^\circ, 180^\circ]$ ensures $y_0 > 0.7112 \text{ m}$ and $x_0 < 0$.
- Set the motor to release at ϕ , where $\theta = \phi - 180^\circ$.
- Adjust v_0 via motor speed based on d and θ .
- Use numerical methods for precise v_0 calculations.